Exercise 9G — Worked Solutions

Question 1

For each of the following, solve the equation over $\mathbb{C}$ and show the solutions on an Argand diagram.

a Solve $z^2 + 1 = 0$.

Solution

$z^2 + 1 = 0$ gives $z^2 = -1$. Let $z = r\,\mathrm{cis}\,\theta$. Note that $-1 = \mathrm{cis}\,(\pi)$.

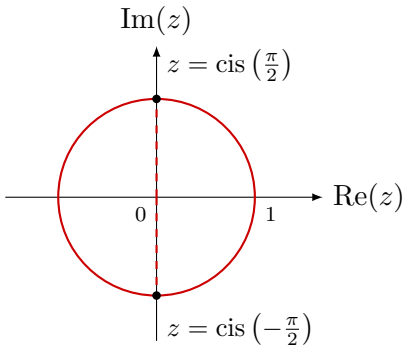
$$\begin{aligned} & (r\,\mathrm{cis}\,\theta)^2 = 1\,\mathrm{cis}\,(\pi) \\ \therefore \quad & r^2\,\mathrm{cis}(2\theta) = 1\,\mathrm{cis}\,(\pi) \\ \therefore \quad & r^2 = 1 \quad \text{and} \quad 2\theta = \pi + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad & r = 1 \quad \text{and} \quad \theta = \frac{\pi}{2} + k\pi \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

Hence $z = \mathrm{cis}\left(\frac{\pi}{2} + k\pi\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = \mathrm{cis}\left(\frac{\pi}{2}\right)$

For $k = 1$: $z = \mathrm{cis}\left(\frac{3\pi}{2}\right) = \mathrm{cis}\left(-\frac{\pi}{2}\right)$

The two solutions are $\mathrm{cis}\left(\frac{\pi}{2}\right)$ and $\mathrm{cis}\left(-\frac{\pi}{2}\right)$.
They lie on the unit circle at intervals of π around the circle.



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b Solve $z^3 = 27i$.

Solution

Let $z = r\,\mathrm{cis}\,\theta$. Note that $27i = 27\,\mathrm{cis}\left(\frac{\pi}{2}\right)$.

$$\begin{aligned} & (r\,\mathrm{cis}\,\theta)^3 = 27\,\mathrm{cis}\left(\frac{\pi}{2}\right) \\ \therefore \quad & r^3\,\mathrm{cis}(3\theta) = 27\,\mathrm{cis}\left(\frac{\pi}{2}\right) \\ \therefore \quad & r^3 = 27 \quad \text{and} \quad 3\theta = \frac{\pi}{2} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad & r = 3 \quad \text{and} \quad \theta = \frac{\pi}{6} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

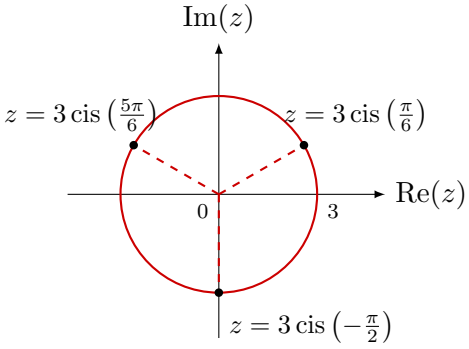
Hence $z = 3\,\mathrm{cis}\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = 3\,\mathrm{cis}\left(\frac{\pi}{6}\right)$

For $k = 1$: $z = 3\,\mathrm{cis}\left(\frac{5\pi}{6}\right)$

For $k = 2$: $z = 3\,\mathrm{cis}\left(\frac{3\pi}{2}\right) = 3\,\mathrm{cis}\left(-\frac{\pi}{2}\right)$

The three solutions are $3\,\mathrm{cis}\left(\frac{\pi}{6}\right)$, $3\,\mathrm{cis}\left(\frac{5\pi}{6}\right)$ and $3\,\mathrm{cis}\left(-\frac{\pi}{2}\right)$.
They lie on a circle of radius 3 at intervals of $\frac{2\pi}{3}$ around the circle.



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c Solve $z^2 = 1 + \sqrt{3}i$.

Solution

Let $z = r\operatorname{cis}\theta$. Note that $1 + \sqrt{3}i = 2\operatorname{cis}\left(\frac{\pi}{3}\right)$, since $|1 + \sqrt{3}i| = 2$ and $\operatorname{Arg}(1 + \sqrt{3}i) = \frac{\pi}{3}$.

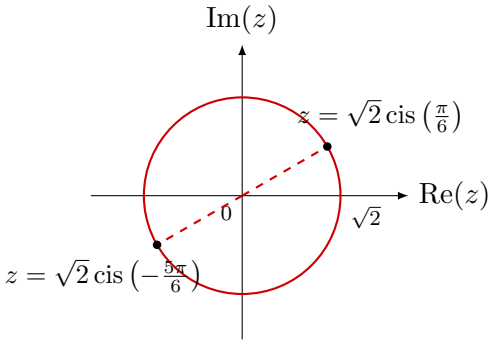
$$\begin{aligned} (r\operatorname{cis}\theta)^2 &= 2\operatorname{cis}\left(\frac{\pi}{3}\right) \\ \therefore \quad r^2\operatorname{cis}(2\theta) &= 2\operatorname{cis}\left(\frac{\pi}{3}\right) \\ \therefore \quad r^2 = 2 \quad &\text{and} \quad 2\theta = \frac{\pi}{3} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = \sqrt{2} \quad &\text{and} \quad \theta = \frac{\pi}{6} + k\pi \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

Hence $z = \sqrt{2}\operatorname{cis}\left(\frac{\pi}{6} + k\pi\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = \sqrt{2}\operatorname{cis}\left(\frac{\pi}{6}\right)$

For $k = 1$: $z = \sqrt{2}\operatorname{cis}\left(\frac{7\pi}{6}\right) = \sqrt{2}\operatorname{cis}\left(-\frac{5\pi}{6}\right)$

The two solutions are $\sqrt{2}\operatorname{cis}\left(\frac{\pi}{6}\right)$ and $\sqrt{2}\operatorname{cis}\left(-\frac{5\pi}{6}\right)$.
They lie on a circle of radius $\sqrt{2}$ at intervals of π around the circle.



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d Solve $z^2 = 1 - \sqrt{3}i$.

Solution

Let $z = r\operatorname{cis}\theta$. Note that $1 - \sqrt{3}i = 2\operatorname{cis}\left(-\frac{\pi}{3}\right)$, since $|1 - \sqrt{3}i| = 2$ and $\operatorname{Arg}(1 - \sqrt{3}i) = -\frac{\pi}{3}$.

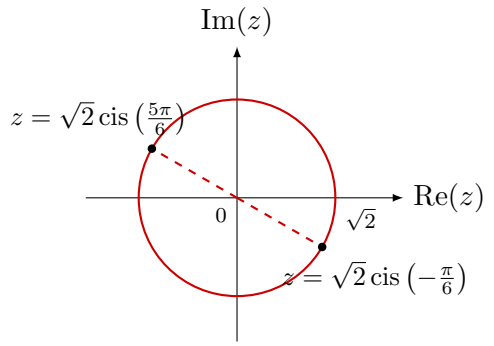
$$\begin{aligned} (r\operatorname{cis}\theta)^2 &= 2\operatorname{cis}\left(-\frac{\pi}{3}\right) \\ \therefore \quad r^2\operatorname{cis}(2\theta) &= 2\operatorname{cis}\left(-\frac{\pi}{3}\right) \\ \therefore \quad r^2 = 2 \quad &\text{and} \quad 2\theta = -\frac{\pi}{3} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = \sqrt{2} \quad &\text{and} \quad \theta = -\frac{\pi}{6} + k\pi \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

Hence $z = \sqrt{2}\operatorname{cis}\left(-\frac{\pi}{6} + k\pi\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = \sqrt{2}\operatorname{cis}\left(-\frac{\pi}{6}\right)$

For $k = 1$: $z = \sqrt{2}\operatorname{cis}\left(\frac{5\pi}{6}\right)$

The two solutions are $\sqrt{2}\operatorname{cis}\left(-\frac{\pi}{6}\right)$ and $\sqrt{2}\operatorname{cis}\left(\frac{5\pi}{6}\right)$.
They lie on a circle of radius $\sqrt{2}$ at intervals of π around the circle.



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e Solve $z^3 = i$.

Solution

Let $z = r\,\mathrm{cis}\,\theta$. Note that $i = \mathrm{cis}\left(\frac{\pi}{2}\right)$.

$$\begin{aligned} (r\,\mathrm{cis}\,\theta)^3 &= 1\,\mathrm{cis}\left(\frac{\pi}{2}\right) \\ \therefore \quad r^3\,\mathrm{cis}(3\theta) &= 1\,\mathrm{cis}\left(\frac{\pi}{2}\right) \\ \therefore \quad r^3 = 1 \quad \text{and} \quad 3\theta &= \frac{\pi}{2} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = 1 \quad \text{and} \quad \theta &= \frac{\pi}{6} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

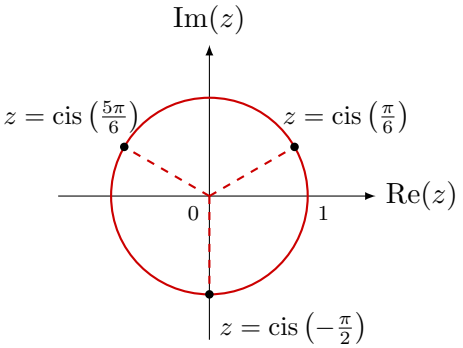
Hence $z = \mathrm{cis}\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = \mathrm{cis}\left(\frac{\pi}{6}\right)$

For $k = 1$: $z = \mathrm{cis}\left(\frac{5\pi}{6}\right)$

For $k = 2$: $z = \mathrm{cis}\left(\frac{3\pi}{2}\right) = \mathrm{cis}\left(-\frac{\pi}{2}\right)$

The three solutions are $\mathrm{cis}\left(\frac{\pi}{6}\right)$, $\mathrm{cis}\left(\frac{5\pi}{6}\right)$ and $\mathrm{cis}\left(-\frac{\pi}{2}\right)$. They lie on the unit circle at intervals of $\frac{2\pi}{3}$ around the circle.



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f Solve $z^3 + i = 0$.

Solution

$z^3 + i = 0$ gives $z^3 = -i$. Let $z = r\,\mathrm{cis}\,\theta$. Note that $-i = \mathrm{cis}\left(-\frac{\pi}{2}\right)$.

$$\begin{aligned} (r\,\mathrm{cis}\,\theta)^3 &= 1\,\mathrm{cis}\left(-\frac{\pi}{2}\right) \\ \therefore \quad r^3\,\mathrm{cis}(3\theta) &= 1\,\mathrm{cis}\left(-\frac{\pi}{2}\right) \\ \therefore \quad r^3 = 1 \quad \text{and} \quad 3\theta &= -\frac{\pi}{2} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = 1 \quad \text{and} \quad \theta &= -\frac{\pi}{6} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

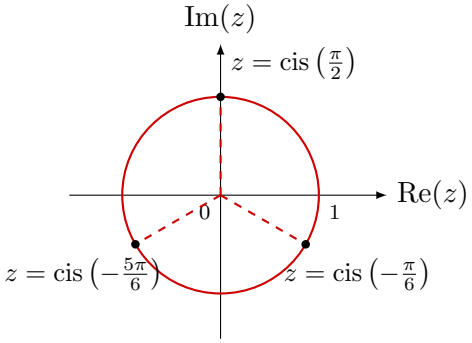
Hence $z = \mathrm{cis}\left(-\frac{\pi}{6} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = \mathrm{cis}\left(-\frac{\pi}{6}\right)$

For $k = 1$: $z = \mathrm{cis}\left(\frac{\pi}{2}\right)$

For $k = 2$: $z = \mathrm{cis}\left(\frac{7\pi}{6}\right) = \mathrm{cis}\left(-\frac{5\pi}{6}\right)$

The three solutions are $\mathrm{cis}\left(-\frac{\pi}{6}\right)$, $\mathrm{cis}\left(\frac{\pi}{2}\right)$ and $\mathrm{cis}\left(-\frac{5\pi}{6}\right)$. They lie on the unit circle at intervals of $\frac{2\pi}{3}$ around the circle.



Question 2

Find all the cube roots of the following complex numbers.

a Find the cube roots of $4\sqrt{2} - 4\sqrt{2}i$.

Solution

Let $z = r\,\mathrm{cis}\,\theta$ be a cube root of $4\sqrt{2}-4\sqrt{2}i$, so that $z^3 = 4\sqrt{2}-4\sqrt{2}i$. Note that $4\sqrt{2}-4\sqrt{2}i = 8\,\mathrm{cis}\left(-\frac{\pi}{4}\right)$, since $|4\sqrt{2} - 4\sqrt{2}i| = 8$ and $\mathrm{Arg}(4\sqrt{2} - 4\sqrt{2}i) = -\frac{\pi}{4}$.

$$
\begin{aligned}
(r\,\mathrm{cis}\,\theta)^3 &= 8\,\mathrm{cis}\left(-\frac{\pi}{4}\right) \\
\therefore \quad r^3\,\mathrm{cis}(3\theta) &= 8\,\mathrm{cis}\left(-\frac{\pi}{4}\right) \\
\therefore \quad r^3 = 8 \quad &\text{and} \quad 3\theta = -\frac{\pi}{4} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\
\therefore \quad r = 2 \quad &\text{and} \quad \theta = -\frac{\pi}{12} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z}
\end{aligned}
$$

Hence $z = 2\,\mathrm{cis}\left(-\frac{\pi}{12} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

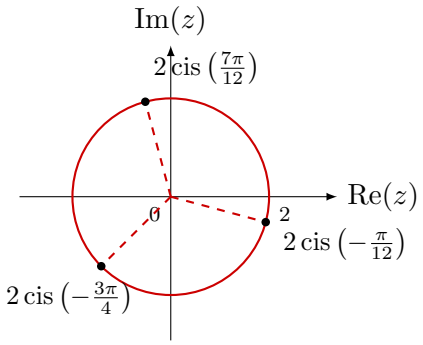
For $k = 0$: $z = 2\,\mathrm{cis}\left(-\frac{\pi}{12}\right)$

For $k = 1$: $z = 2\,\mathrm{cis}\left(\frac{7\pi}{12}\right)$

For $k = 2$: $z = 2\,\mathrm{cis}\left(\frac{5\pi}{4}\right) = 2\,\mathrm{cis}\left(-\frac{3\pi}{4}\right)$

The three solutions are $2\,\mathrm{cis}\left(-\frac{\pi}{12}\right)$, $2\,\mathrm{cis}\left(\frac{7\pi}{12}\right)$ and $2\,\mathrm{cis}\left(-\frac{3\pi}{4}\right)$.

They lie on a circle of radius 2 at intervals of $\frac{2\pi}{3}$ around the circle.



b Find the cube roots of $-4\sqrt{2}+4\sqrt{2}i$.

Solution

Let $z = r\operatorname{cis}\theta$ be a cube root of $-4\sqrt{2}+4\sqrt{2}i$, so that $z^3 = -4\sqrt{2}+4\sqrt{2}i$. Note that $-4\sqrt{2}+4\sqrt{2}i = 8\operatorname{cis}\left(\frac{3\pi}{4}\right)$, since $|-4\sqrt{2}+4\sqrt{2}i| = 8$ and $\operatorname{Arg}(-4\sqrt{2}+4\sqrt{2}i) = \frac{3\pi}{4}$.

$$\begin{aligned}
(r\operatorname{cis}\theta)^3 &= 8\operatorname{cis}\left(\frac{3\pi}{4}\right)\\
\therefore\quad r^3\operatorname{cis}(3\theta) &= 8\operatorname{cis}\left(\frac{3\pi}{4}\right)\\
\therefore\quad r^3 = 8 \quad\text{and}\quad 3\theta &= \frac{3\pi}{4}+2k\pi \quad\text{where } k\in\mathbb{Z}\\
\therefore\quad r = 2 \quad\text{and}\quad \theta &= \frac{\pi}{4}+\frac{2k\pi}{3} \quad\text{where } k\in\mathbb{Z}
\end{aligned}$$

Hence $z = 2\operatorname{cis}\left(\frac{\pi}{4}+\frac{2k\pi}{3}\right)$, where $k\in\mathbb{Z}$.

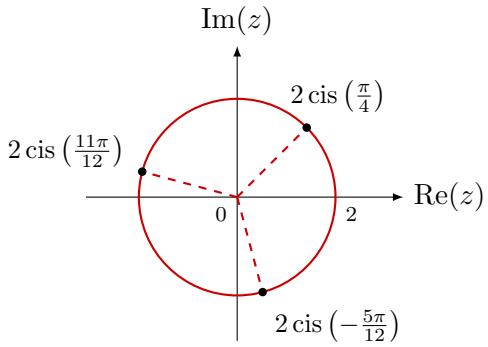
For $k=0$: $z = 2\operatorname{cis}\left(\frac{\pi}{4}\right)$

For $k=1$: $z = 2\operatorname{cis}\left(\frac{11\pi}{12}\right)$

For $k=2$: $z = 2\operatorname{cis}\left(\frac{19\pi}{12}\right) = 2\operatorname{cis}\left(-\frac{5\pi}{12}\right)$

The three solutions are $2\operatorname{cis}\left(\frac{\pi}{4}\right)$, $2\operatorname{cis}\left(\frac{11\pi}{12}\right)$ and $2\operatorname{cis}\left(-\frac{5\pi}{12}\right)$.

They lie on a circle of radius 2 at intervals of $\frac{2\pi}{3}$ around the circle.



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c Find the cube roots of $-4\sqrt{3}-4i$.

Solution

Let $z = r\operatorname{cis}\theta$ be a cube root of $-4\sqrt{3}-4i$, so that $z^3 = -4\sqrt{3}-4i$. Note that $-4\sqrt{3}-4i = 8\operatorname{cis}\left(-\frac{5\pi}{6}\right)$, since $|-4\sqrt{3}-4i| = 8$ and $\operatorname{Arg}(-4\sqrt{3}-4i) = -\frac{5\pi}{6}$.

$$\begin{aligned}
(r\operatorname{cis}\theta)^3 &= 8\operatorname{cis}\left(-\frac{5\pi}{6}\right)\\
\therefore\quad r^3\operatorname{cis}(3\theta) &= 8\operatorname{cis}\left(-\frac{5\pi}{6}\right)\\
\therefore\quad r^3 = 8 \quad\text{and}\quad 3\theta &= -\frac{5\pi}{6}+2k\pi \quad\text{where } k\in\mathbb{Z}\\
\therefore\quad r = 2 \quad\text{and}\quad \theta &= -\frac{5\pi}{18}+\frac{2k\pi}{3} \quad\text{where } k\in\mathbb{Z}
\end{aligned}$$

Hence $z = 2\operatorname{cis}\left(-\frac{5\pi}{18}+\frac{2k\pi}{3}\right)$, where $k\in\mathbb{Z}$.

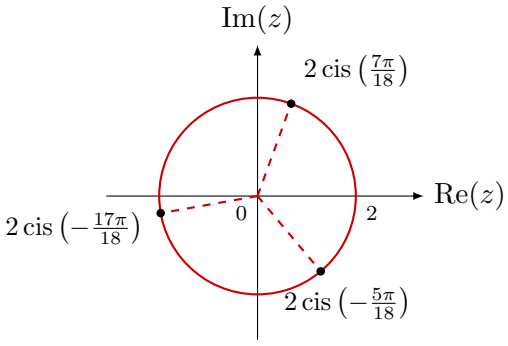
For $k=0$: $z = 2\operatorname{cis}\left(-\frac{5\pi}{18}\right)$

For $k=1$: $z = 2\operatorname{cis}\left(\frac{7\pi}{18}\right)$

For $k=2$: $z = 2\operatorname{cis}\left(\frac{19\pi}{18}\right) = 2\operatorname{cis}\left(-\frac{17\pi}{18}\right)$

The three solutions are $2\operatorname{cis}\left(-\frac{5\pi}{18}\right)$, $2\operatorname{cis}\left(\frac{7\pi}{18}\right)$ and $2\operatorname{cis}\left(-\frac{17\pi}{18}\right)$.

They lie on a circle of radius 2 at intervals of $\frac{2\pi}{3}$ around the circle.



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d Find the cube roots of $4\sqrt{3} - 4i$.

Solution

Let $z = r\,\text{cis}\,\theta$ be a cube root of $4\sqrt{3} - 4i$, so that $z^3 = 4\sqrt{3} - 4i$. Note that $4\sqrt{3} - 4i = 8\,\text{cis}\left(-\frac{\pi}{6}\right)$, since $|4\sqrt{3} - 4i| = 8$ and $\text{Arg}(4\sqrt{3} - 4i) = -\frac{\pi}{6}$.

$$\begin{aligned} (r\,\text{cis}\,\theta)^3 &= 8\,\text{cis}\left(-\frac{\pi}{6}\right) \\ \therefore \quad r^3\,\text{cis}(3\theta) &= 8\,\text{cis}\left(-\frac{\pi}{6}\right) \\ \therefore \quad r^3 = 8 \quad &\text{and} \quad 3\theta = -\frac{\pi}{6} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = 2 \quad &\text{and} \quad \theta = -\frac{\pi}{18} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

Hence $z = 2\,\text{cis}\left(-\frac{\pi}{18} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

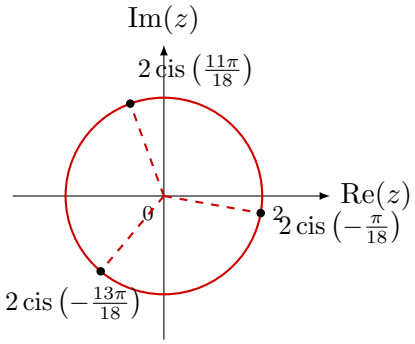
For $k = 0$: $z = 2\,\text{cis}\left(-\frac{\pi}{18}\right)$

For $k = 1$: $z = 2\,\text{cis}\left(\frac{11\pi}{18}\right)$

For $k = 2$: $z = 2\,\text{cis}\left(\frac{23\pi}{18}\right) = 2\,\text{cis}\left(-\frac{13\pi}{18}\right)$

The three solutions are $2\,\text{cis}\left(-\frac{\pi}{18}\right)$, $2\,\text{cis}\left(\frac{11\pi}{18}\right)$ and $2\,\text{cis}\left(-\frac{13\pi}{18}\right)$.

They lie on a circle of radius 2 at intervals of $\frac{2\pi}{3}$ around the circle.



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e Find the cube roots of $-125i$.

Solution

Let $z = r\,\text{cis}\,\theta$ be a cube root of $-125i$, so that $z^3 = -125i$. Note that $-125i = 125\,\text{cis}\left(-\frac{\pi}{2}\right)$, since $|-125i| = 125$ and $\text{Arg}(-125i) = -\frac{\pi}{2}$.

$$\begin{aligned} (r\,\text{cis}\,\theta)^3 &= 125\,\text{cis}\left(-\frac{\pi}{2}\right) \\ \therefore \quad r^3\,\text{cis}(3\theta) &= 125\,\text{cis}\left(-\frac{\pi}{2}\right) \\ \therefore \quad r^3 = 125 \quad &\text{and} \quad 3\theta = -\frac{\pi}{2} + 2k\pi \quad \text{where } k \in \mathbb{Z} \\ \therefore \quad r = 5 \quad &\text{and} \quad \theta = -\frac{\pi}{6} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z} \end{aligned}$$

Hence $z = 5\,\text{cis}\left(-\frac{\pi}{6} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

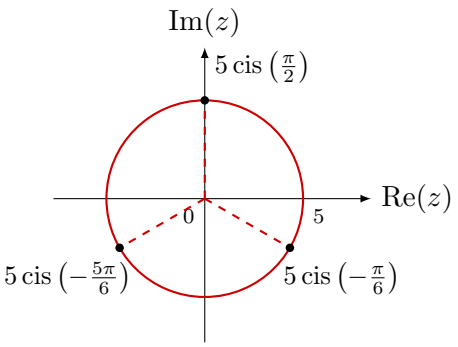
For $k = 0$: $z = 5\,\text{cis}\left(-\frac{\pi}{6}\right)$

For $k = 1$: $z = 5\,\text{cis}\left(\frac{\pi}{2}\right)$

For $k = 2$: $z = 5\,\text{cis}\left(\frac{7\pi}{6}\right) = 5\,\text{cis}\left(-\frac{5\pi}{6}\right)$

The three solutions are $5\,\text{cis}\left(-\frac{\pi}{6}\right)$, $5\,\text{cis}\left(\frac{\pi}{2}\right)$ and $5\,\text{cis}\left(-\frac{5\pi}{6}\right)$.

They lie on a circle of radius 5 at intervals of $\frac{2\pi}{3}$ around the circle.



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f Find the cube roots of $-1+i$.

Solution

Let $z = r\,\mathrm{cis}\,\theta$ be a cube root of $-1+i$, so that $z^3 = -1+i$. Note that $-1+i = \sqrt{2}\,\mathrm{cis}\left(\frac{3\pi}{4}\right)$, since $|-1+i| = \sqrt{2}$ and $\mathrm{Arg}(-1+i) = \frac{3\pi}{4}$.

$$(r\,\mathrm{cis}\,\theta)^3 = \sqrt{2}\,\mathrm{cis}\left(\frac{3\pi}{4}\right)$$

$$\therefore \quad r^3\,\mathrm{cis}(3\theta) = 2^{\frac{1}{2}}\,\mathrm{cis}\left(\frac{3\pi}{4}\right)$$

$$\therefore \quad r^3 = 2^{\frac{1}{2}} \quad \text{and} \quad 3\theta = \frac{3\pi}{4} + 2k\pi \quad \text{where } k \in \mathbb{Z}$$

$$\therefore \quad r = 2^{\frac{1}{6}} \quad \text{and} \quad \theta = \frac{\pi}{4} + \frac{2k\pi}{3} \quad \text{where } k \in \mathbb{Z}$$

Hence $z = 2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{\pi}{4} + \frac{2k\pi}{3}\right)$, where $k \in \mathbb{Z}$.

For $k = 0$: $z = 2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{\pi}{4}\right)$

For $k = 1$: $z = 2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{11\pi}{12}\right)$

For $k = 2$: $z = 2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{19\pi}{12}\right) = 2^{\frac{1}{6}}\,\mathrm{cis}\left(-\frac{5\pi}{12}\right)$

The three solutions are $2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{\pi}{4}\right)$, $2^{\frac{1}{6}}\,\mathrm{cis}\left(\frac{11\pi}{12}\right)$ and $2^{\frac{1}{6}}\,\mathrm{cis}\left(-\frac{5\pi}{12}\right)$.

They lie on a circle of radius $2^{\frac{1}{6}}$ at intervals of $\frac{2\pi}{3}$ around the circle.

